

4 (a) force is proportional to extension B1 [1]

(b) (i) gradient of graph determined (e.g. $50 / 40 \times 10^{-3}$) = 1250 N m^{-1} A1 [1]

(ii) $W = \frac{1}{2} k x^2$ or $W = \frac{1}{2} \text{ final force} \times \text{extension}$ M1
 $= 0.5 \times 1250 \times (36 \times 10^{-3})^2$ or $0.5 \times 45 \times 36 \times 10^{-3}$ M1
 $= 0.81 \text{ J}$ A0 [2]

(c) (i) $0.81 = \frac{1}{2} m v^2$ C1
 $v = 8.0 \text{ (8.0498) m s}^{-1}$ A1 [2]

(ii) $4 \times \text{KE} / 4 \times \text{WD}$ or 3.24 J C1
hence twice the compression = 72 mm A1 [2]

(iii) Max height is when all KE or WD C1
or elastic PE is converted to GPE
ratio = $1/4$ or 0.25 A1 [2]

5 (a) (i) Start from (0,0) and smooth curve in correct direction B1